



June 2015

Team: SC, RM, LC (75%), NW (20%)

Strategy

- [LC] Research and reporting on landscape of primary mortgage market
 - Weighted Average Incentive of the Mortgage Universe: Both pool- and loan-level calculations
 - Different variations on the % Refinanceable Index: %Borrowers who've seen a low in rates for the first, rate sensitivity of these borrowers. Productionized this report.
 - Productionized report that summarizes mortgage rate differences between large and small lenders
- [RM, LC] Mortgage Credit Availability Index: Use Default forecasts from Accordion to track the credit quality of Purchase originations over time.
- [SC] CRE Business Launch: DDQ for Equity Long/Short strategy
- [SC] Refinement of IOS Basis Model

Modeling

- **Scale:** Reorganize and refactor existing CCMQuantlib fixed income analytics library (written by Bin Shao) as Scale. See the Scale presentation (distributed separately) for an overarching view of the goals of this library.
 - [JH, NW, RM] New Scale: Work on productionizing Quantlib code that replicates Bin's code. If successful, this approach would provide an excellent platform for our Scale project. Quantlib is:
 - Reasonably well-tested: it's been around since 2000
 - Well-maintained: There are at least a couple of releases every year
 - Rich feature set: Quantlib functionality includes a significant collection of utilities (Date libraries, optimization engines, splines etc.), yield curve builders, interest rate models etc.
 - Documented code
 - [NW] Fixed various errors in the Yield Curve Builder and now able to build it from 01/02/2014 every day without checking estimates against input.
- **Baton** [RM]: Agency prepayment model



- Completed work on Default sub-model. Waiting to present to Investment Committee.
- Started work on Turnover sub-model.

Analytics

- **Ensemble** [RM, JH]: Middle-tier – allows for decoupling Scale from front-end applications
 - Expose functions in Scale for front-end users to test/use the Scale library: Completed work on this. Daily build in production.



July 2015 Pipeline

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Strategy

- [LC] Research and reporting on landscape of primary mortgage market
 - Best execution
 - Productionize weighted-average incentive calculation
- [LC, RM] First version of proprietary Mortgage Credit Availability (MCA) measure by running the most recent originations through Accordion each month and tracking a default measure.
- [SC] Refinement of IOS Basis Model
- [SC] CRE Strategy: Document Investment Process and Risk Management strategy
- [SC, LC] Slides for Quarterly Call

Modeling

- **Scale** [NW] Write native code instead of relying on QuantLib itself for checking results produced by Yield Curve Builder. [RM] Add logging structure to Scale Prepayment Model. [NW, RM] Add Unit Test framework to Yield Curve Builder, Prepayment Model.
- **Accordion** [RM, LC]: Roll-rate model for GSE loan-level data
 - Productionize running CRT/STACR deals with eMBS CRT data on a monthly basis [July]
 - Create a monthly surveillance report based on the CRT monthly files [July]
 - Generate a daily forecast of HPA/mortgage rates for the base case.
 - Generate daily delinquency/default vectors for CRT deals using stress scenarios from Moody's/Economy.com and a base case.
 - Generate daily yield numbers for various CRT deals and tranches using these delinquency/default vectors.
 - Create a monthly surveillance report based on the CRT monthly files
- **Baton** [RM]: Agency prepayment-model
 - Turnover sub-model